

Minsuk Kim

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CONTACT INFORMATION	Senseable City Lab Department of Urban Studies and Planning Massachusetts Institute of Technology 77 Massachusetts Avenue, Cambridge, MA 02139	dan418@mit.edu Google Scholar ORCID minsuk-daniel-kim.me
RESEARCH STATEMENT	My research explores the interplay between transportation infrastructure, human mobility, and the accessibility of urban amenities. I employ tools from network science, statistical physics, and machine learning to uncover the underlying dynamics that shape mobility patterns and urban accessibility. My work extends to broader topics such as phase transitions and dynamics on networks, unsupervised representation learning, and geospatial data analysis with the goal of advancing our understanding of various complex systems.	
APPOINTMENT	Massachusetts Institute of Technology , Cambridge, MA, USA • Postdoctoral Fellow, Senseable City Lab	June 2026 –current
EDUCATION	Ph.D. Informatics, Indiana University, Bloomington, IN, USA • Track: Complex Networks and Systems • Minor: Computer Science • Committee: Filippo Radicchi (Chair), Santo Fortunato , Haewoon Kwak , and Roni Khardon • Dissertation: Data-driven modeling of complex transportation and mobility systems M.S. Physics, Kyung Hee University, Seoul, Korea • Concentration: Statistical Physics • Advisor: Professor Soon-Hyung Yook • Thesis: Spreading of consensus and epidemic on complex networks (Written in Korean) B.S. Physics, Kyung Hee University, Seoul, Korea	August 2022–May 2026 March 2019–February 2021 March 2013–February 2019
PUBLICATIONS	Journal Articles J1. Kim, Minsuk , Cirigliano, L., Castellano, C., Sun, H., Jankowski, R., Poggialini, A. & Radicchi, F. Shortest-path percolation on scale-free networks. <i>Physical Review E</i> 113 , 014314 (2026). J2. Kim, Minsuk , Diggans, C. T. & Radicchi, F. Modeling resource consumption in the US air transportation system via minimum-cost percolation. <i>Nature Communications</i> 16 , 8105 (2025). J3. Kim, Minsuk & Yook, S.-H. Effect of isolation strategies on epidemic spreading in multiplex networks with an annealed interactions layer. <i>Journal of the Korean Physical Society</i> 86 , 73–84 (2025). J4. Kim, Minsuk & Radicchi, F. Shortest-Path Percolation on Random Networks. <i>Physical Review Letters</i> 133 , 047402 (2024). J5. Kaiser, D., Patwardhan, S., Kim, Minsuk & Radicchi, F. Reconstruction of multiplex networks via graph embeddings. <i>Physical Review E</i> 109 , 024313 (2024). J6. Kim, Minsuk & Yook, S.-H. Majority-vote model with degree-weighted influence on complex networks. <i>Physical Review E</i> 103 , 022302 (2021). Peer-reviewed Conference Proceedings	

- C1. Aiyappa, R., DeVerna, M. R., Pote, M., Truong, B. T., Zhao, W., Axelrod, D., Pessianzadeh, A., Kachwala, Z., Kim, M., Seckin, O. C., **Kim, Minsuk**, Gandhi, S., Manikonda, A., Pierri, F., Menczer, F. & Yang, K.-C. *A multi-platform collection of social media posts about the 2022 us midterm elections in Proceedings of the international AAAI conference on web and social media* **17** (2023), 981–989.

Under Review

- R1. Weng, G., **Kim, Minsuk**, Ahn, Y.-Y. & Moro, E. Beyond Distance: Mobility Neural Embeddings Reveal Visible and Invisible Barriers in Urban Space. *arXiv:2506.24061* (2025).

HONORS AND AWARDS

Fellowship

- Luddy Excellence Award, Indiana University, 2026
- U.S. Department of the Air Force Summer Faculty Fellowship Program, 2024
- Luddy Doctoral Summer Fellowship, Indiana University, 2023, 2025

Scholarship

- Scholarship of Dean of Graduate School, Kyung Hee University, 2019
- Superiority Scholarship, Kyung Hee University, 2017
- Exemplary Scholarship, Kyung Hee University, 2013, 2014

Conference

- Best Oral Presentation Award, The 21st Workshop for Statistical Physics (Online), 2021
- Best Poster Presentation Award, The Korean Physical Society Spring Meeting, (Online), 2020
- Best Oral Presentation Award, The 20th Workshop for Statistical Physics (Byeonsan, Korea), 2019

Miscellaneous

- Best Project Award, The 19th KIAS-APCTP Winter School on Statistical Physics, (Pohang, Korea), 2022
- Best Project Award, The 17th KIAS-APCTP Winter School on Statistical Physics, (Pohang, Korea), 2020
- Excellence Project Award (3rd place), The 10th KIAS CAC Summer School on Scientific Computing & Artificial Intelligence, (Seoul, Korea), 2019

TEACHING

Indiana University

- Associate Instructor
 - Performance Analytics (INFO-I369) Spring 2025, Fall 2025

Kyung Hee University

- Guest Lecturer
 - Advanced Statistical Physics (PHYS7037) November 15th, 2021
title: “Finite-size scaling analysis in Statistical Physics”
 - Statistical Mechanics 2 (PHYS7072) December 14th, 2020
title: “Statistical Physics of Opinion Dynamics”
- Teaching Assistant
 - Modern Physics (PHYS2309) Spring 2019, Spring 2020
 - Computational Physics (PHYS3308) Spring 2019, Spring 2020
 - Informations for Physics (PHYS3310) Fall 2019, Fall 2020

MENTORING	M.S. and post-undergraduate students	
	Bilal Sultan (Data Science, Indiana University)	2025-present
	Undergraduate students	
	Bilal Sultan (Data Science, Indiana University)	2024-2025
PRESENTATIONS	Conference talks	
	- NetSci 2026, “Shortest-path percolation on scale-free networks” (Boston, IN, USA, June, 2026) - NetSci 2025, “Minimum-cost percolation on US air transportation network” (Maastricht, Netherlands, June, 2025) - NetSci 2025 Satellite Symposium: Critical Phenomena In Networks, “Shortest-path percolation on complex networks” (Maastricht, Netherlands, June, 2025) - IC2S2 2024, “Unsupervised embedding of mobility reveals invisible barriers in US cities” (Philadelphia, PA, USA, July, 2024) - NetSci 2024, “Shortest-path percolation on complex networks” (Québec City, Canada, June, 2024) - APS March Meeting, “Shortest-path percolation on complex networks” (Minneapolis, MN, USA, March, 2024) - The 21st Workshop for Statistical Physics, “Control of epidemic spreading on layered networks” (Online, August, 2021) - KPS Spring Meeting, “Effective epidemic control model on multilayer networks” (Online, April, 2021) - The 20th Workshop for Statistical Physics, “Degree-weighted Majority-vote model on Complex Networks” (Byeonsan, Korea, August, 2019)	
	Conference posters	
	- NetMob 2024 data challenge, “Investigating Human Mobility Patterns with Field Theory and Open Data Sources” (Washington, D.C., USA, October, 2024) - KPS Spring Meeting, “Impact of degree-weighted influence in the Majority-vote dynamics on Complex Networks” (Online, July, 2020) - KPS Fall Meeting, “Majority-vote model with degree-weighted influence on Complex Networks” (Gwangju, Korea, October, 2019)	
	Colloquia and Seminars	
	- Department of Physics, Kyung Hee University, “Shortest-path percolation on complex networks”, (Seoul, Korea, December 20th, 2024) Complex Systems Research Exchange (CREx), “Shortest-path percolation on complex networks”, (Online, January 30th, 2024) - Kyung Hee University Academic Society Seminar: Scaling and Theoretical Physics, “Finite-size scaling analysis in Statistical Physics” (Seoul, Korea, November 26th, 2021) - Kyung Hee University Theoretical physics research group seminar, “Majority-vote model with degree influence on complex networks” (Seoul, Korea, May 22nd, 2020)	
SERVICE	Reviewer	
	- Journal: Communications Physics, IEEE Transaction on Network Science and Engineering, Nature Communications, Physical Review E, Scientific Reports - Conference: NetSci (2026, 2025), IC2S2 (2025, 2024)	
SKILLS	Programming Language: C, Python, R	

Relevant tools: NetworkX, Gephi, OSMnx, Geopandas, statsmodels

Specialization: Monte Carlo simulation, Finite-size scaling analysis

Language: Korean (native), English (fluent)

REFERENCES

Dr. Filippo Radicchi

Professor

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Dr. Yong-Yeol Ahn

Quantitative Foundation Distinguished Professor

School of Data Science

University of Virginia

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Dr. Esteban Moro

Professor

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Northeastern University

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Dr. Santo Fortunato

James H. Rudy Professor of Informatics and Computing

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Dr. Soon-Hyung Yook

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